Problem-Solving & Data

Section 1: Rates, Proportions & Dimensional Analysis

1 A car travels at a constant speed of 66 feet per second. What is this speed in miles per hour? (1 mile $= 5,280$ feet)

a. 40 mph

b. 45 mph

c. 50 mph

d. 55 mph

2 On a map, 2.5 centimeters represents an actual distance of 40 kilometers. If two cities are 6.5 centimeters apart on the map, what is the actual distance between them?

a. 96 km

b. 104 km

c. 112 km

d. 120 km

3 A water hose fills a 120-gallon tank in 15 minutes. At this rate, how many gallons of water does the hose pump in 2 hours?

a. 720

b. 840

c. 960

d. $1,080$

Section 2: Percentages & Successive Changes

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4 The price of an item was originally 80. It was increased by 20% and then the new price was discounted by 10%. What is the final price of the item?

a. 86.40

b. 88.00

c. 90.20

d. 92.00

5 A store manager discounts a winter jacket by 30%, making the sale price 84. What was the original price of the jacket?

a. 110

b. 115

c. 120

d. 125

6 In a chemistry lab, a 400-milliliter solution is 15% acid. How many milliliters of pure water must be added to dilute the solution to 10% acid?

a. 150 mL

b. 200 mL

c. 250 mL

d. 300 mL

Section 3: Scatterplots & Lines of Best Fit

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7 The line of best fit for a scatterplot relating weekly study hours x to exam score y is given by $\hat{y} = 4.5x + 52$. What is the predicted exam score for a student who studies 8 hours?

a. 84

b. 86

c. 88

d. 90

8 For the model $\hat{y} = 3x + 25$, a student with $x = 6$ had an actual exam score of 47. What is the residual for this data point?

a. $+4$

b. -4

c. $+2$

d. -2

9 In the linear regression equation $\hat{y} = -2.4x + 68$, where x is the temperature in degrees Celsius and y is the heating cost in dollars, what does the slope -2.4 represent?

a. For each $1\,^\circ\text{C}$ increase in temperature, heating cost decreases by 2.40 dollars.

b. For each $1\,^\circ\text{C}$ increase in temperature, heating cost increases by 2.40 dollars.

c. The heating cost when temperature is $0\,^\circ\text{C}$ is 2.40 dollars.

d. The maximum possible temperature is $28.3\,^\circ\text{C}$.

Section 4: Data Tables & Proportions

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10 A company surveys 200 employees regarding remote work. If 120 full-time employees and 80 part-time employees participated, and 65% of full-time and 40% of part-time employees favor remote work, what percentage of all surveyed employees favor remote work?

a. 52.5%

b. 55.0%

c. 57.5%

d. 60.0%

11 The ratio of boys to girls in a math competition is $5 : 7$. If there are 35 boys, how many total students are in the competition?

a. 70

b. 84

c. 91

d. 98

12 If x is inversely proportional to y, and $x = 12$ when $y = 6$, what is the value of y when $x = 8$?

a. 4

b. 9

c. 12

d. 16

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